VARIANCE-SENSITIVE MATRIX SPENCER

SUVRIT SRA*

ABSTRACT. This note records a statement refuted in the *Count ex Machina* archive: each is quoted as posed and answered by an explicit witness, and the archive carries a certificate that recomputes the refutation by exact arithmetic.

Summary: Variance-sensitive Matrix Spencer for n contractions of size $n \times n$ (Bandeira, 2016)
Certificate: Exact $d = n$ diagonal family: $\mathrm{disc} = (m-1)/\sqrt{m}$ against variance $1 + 1/m$, a ratio of $\Theta(\sqrt{\log n})$.

Posed by A. S. Bandeira (2016); counterexample found by GPT-5.5 (Pro) (2026-05-24); formalized by E. Akbaş and S. Sra (2026); audited by S. Sra; contributed by S. Sra.

For symmetric matrices $A_1, \ldots, A_n$ the *discrepancy* is $\mathrm{disc}(A_1, \ldots, A_n) = \min_{x \in \{\pm 1\}^n} \| \sum_{i=1}^n x_i A_i \|_{\mathrm{op}}$. Spencer's theorem gives an $O(\sqrt{n})$ bound in the scalar case (and thus also for the commuting matrix case); the Matrix Spencer conjecture asks for the same $O(\sqrt{n})$ bound for n symmetric contractions of size $n \times n$. The variance-sensitive strengthening replaces the ambient scale $\sqrt{n}$ by the intrinsic variance parameter $\| \sum_{i=1}^n A_i^2 \|_{\mathrm{op}}^{1/2}$, which never exceeds $\sqrt{n}$ and is often far smaller; it is the parameter that controls the norm of a random signing in matrix concentration, and Kyng et al. (2020) establish the strengthened bound in the rank-one case.

Problem 1 (As posed: Remark 4.25 of A. S. Bandeira's problem collection (2016).)**.** There exists a universal constant C, such that for any choice of n symmetric matrices $A_1, \ldots, A_n \in \mathbb{R}^{n \times n}$ satisfying $\|A_i\|_{\mathrm{op}} \leq 1$ there exists a coloring $x \in \{\pm 1\}^n$, such that

$$\left\| \sum_{i=1}^n x_i A_i \right\|_{\mathrm{op}} \leq C \left\| \sum_{i=1}^n A_i^2 \right\|_{\mathrm{op}}^{1/2}.$$

The disproof recorded here is not a new contribution of the present manuscript: the family reported below is taken from Theorem A.1 of Akbaş and Sra (2026), the first refutation of the conjecture; but we couldn't resist including it here as an illustration.

Fix an integer $m \geq 2$ and set $n = 2^m$, so that the matrices below are $n \times n$ and there are n of them: the dimension is not inflated relative to the number of summands. Index coordinates by $s = (s_1, \ldots, s_m) \in S := \{\pm 1\}^m$, so $|S| = n$. Write $p = (1, \ldots, 1)$ and $q = (-1, \ldots, -1)$, choose any $U \subseteq S \setminus \{p, q\}$ with $|U| = m$, put $F = S \setminus U$, and fix a bijection $\pi : \{m+1, \ldots, n\} \to F$; note that $p, q \in F$. Define diagonal matrices

$$(A_i)_{s,s} = \frac{s_i}{\sqrt{m}} \quad (1 \leq i \leq m), \qquad A_k = \frac{1}{\sqrt{m}} E_{\pi(k),\pi(k)} \quad (m < k \leq n), \tag{1}$$

where $E_{r,r}$ is the diagonal matrix with a single 1 in coordinate r. Each $\|A_i\|_{\mathrm{op}} = 1/\sqrt{m} \leq 1$, so the family is admissible.

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* TU Munich, Dept. of Mathematics, School of CIT; Garching, Germany. Email: s.sra@tum.de.

From the *Count ex Machina* archive of counterexamples, https://github.com/suvrit/count-ex-machina, whose companion paper is arXiv:2608.29595.

Theorem 2 ([**Disproof**]: variance-sensitive Matrix Spencer)**.** *For the family* (1)*,*

$$\operatorname{disc}(A_1,\dots,A_n)=\frac{m-1}{\sqrt{m}},\qquad \left\|\sum_{i=1}^{n}A_i^2\right\|_{\mathrm{op}}^{1/2}=\sqrt{1+\tfrac{1}{m}},$$

so the ratio of the two sides equals $(m-1)/\sqrt{m+1}=\Theta(\sqrt{\log n})$*. No universal constant* C *can therefore satisfy the variance-sensitive bound, already for* n *matrices of size* $n\times n$*.*

Proof. Since the matrices are diagonal, so is $\sum_i A_i^2$, and its sth entry is

$$\sum_{i=1}^{m}\frac{s_i^2}{m}+\sum_{k=m+1}^{n}\frac{1}{m}\mathbf{1}\{s=\pi(k)\}=1+\frac{1}{m}\mathbf{1}\{s\in F\},$$

whence $\|\sum_i A_i^2\|_{\mathrm{op}}=1+\frac{1}{m}$.

For the discrepancy, let $x\in\{\pm1\}^n$ be an arbitrary signing and $X=\sum_{i=1}^n x_iA_i$, so that

$$X_{s,s}=\frac{1}{\sqrt{m}}\sum_{i=1}^{m}x_is_i+\frac{1}{\sqrt{m}}\mathbf{1}\{s\in F\}x_{\pi^{-1}(s)}.$$

Evaluating at the coordinate $s=(x_1,\dots,x_m)\in S$ makes the first sum equal to m, so

$$\|X\|_{\mathrm{op}}\ge|X_{s,s}|\ge\frac{m}{\sqrt{m}}-\frac{1}{\sqrt{m}}=\frac{m-1}{\sqrt{m}}.$$

This bound is attained. Take $x_1=\cdots=x_m=1$ and $x_{k_p}=-1$, $x_{k_q}=1$, where $k_p=\pi^{-1}(p)$ and $k_q=\pi^{-1}(q)$, with the remaining signs arbitrary. Then $\sqrt{m}X_{s,s}=\sum_{j=1}^m s_j+\mathbf{1}\{s\in F\}x_{\pi^{-1}(s)}$, which equals $m-1$ at $s=p$ and $-(m-1)$ at $s=q$. For every other s the vector is neither all-plus nor all-minus, so $|\sum_j s_j|\le m-2$ and hence $|\sqrt{m}X_{s,s}|\le m-1$. Thus $\|X\|_{\mathrm{op}}=(m-1)/\sqrt{m}$, and the infimum over signings is exactly that.

Dividing the two displayed quantities gives $(m-1)/\sqrt{m+1}$, which grows without bound as $m=\log_2 n\to\infty$. □

Code and certificate

The case is the directory `counterexamples/variance-only-matrix-discrepancy/` of the archive, at commit `73b416e`. Its `verify.py` recomputes every refutation in this note using exact arithmetic or rigorous interval enclosures, never floating-point evidence alone; it raises an error if any check fails and otherwise writes the certificate below. To rerun it:

```
git clone https://github.com/suvrit/count-ex-machina
cd count-ex-machina/counterexamples/variance-only-matrix-discrepancy
python verify.py
```

- Theorem 2 (result `variance-only-matrix-discrepancy`): exact arithmetic.
- `artifacts/certificate.json`: Exact record of the d=n diagonal family: the variance parameter, the exhaustive discrepancy minimum over all 2^n signings, and the ratio identity, for each checked m.

AI usage

The credit line under the title records, for each statement, who posed it, who found its counterexample and when, and who formalized and audited it; the counterexample here was found by GPT-5.5 (Pro) (2026-05-24). This note was assembled mechanically from the case's source in the archive by `tools/notes.py`, written with Claude Code; its mathematics is exactly that of the archive at the commit cited above.

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References

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